

ORF309/MAT380

**Random Walks II: More First Step
Analysis & Gambler's Ruin**

(pp. 122-127)

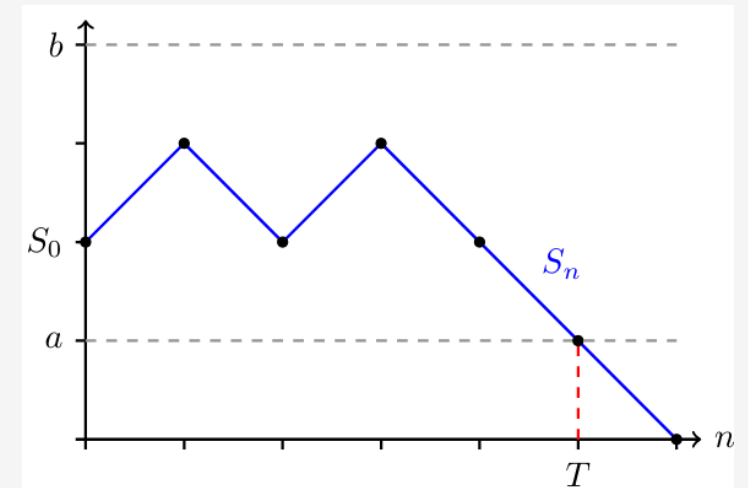
Ioannis Akrotirianakis

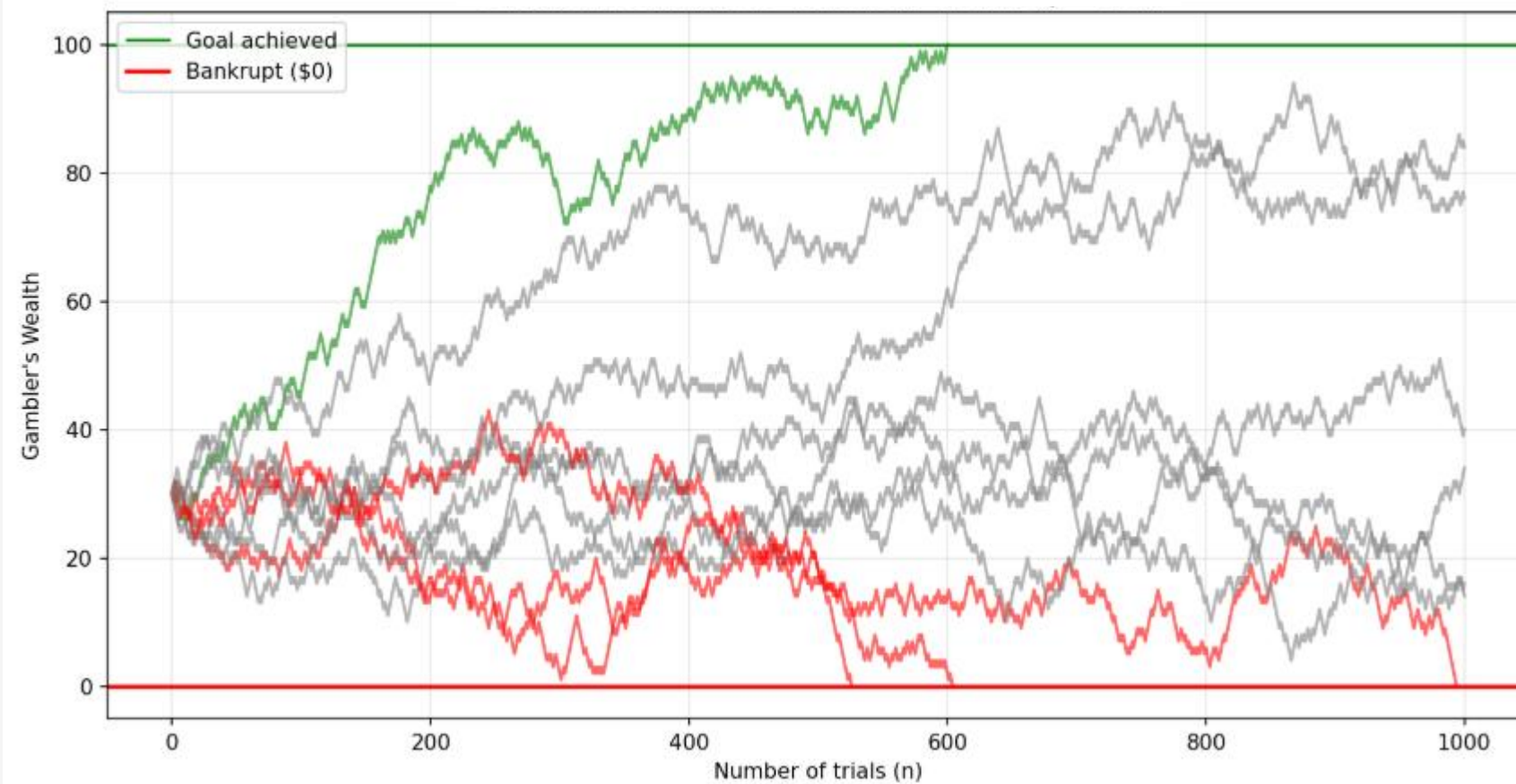
Agenda for today

- Examples
 - **Question 3:** On average, how long will it take for the game to end?
 - **Gambler's ruin**
 - Doubling strategies

Recap

- Consider a **simple symmetric** RW $\{S_n\}_{n \geq 0}$
- Define two **integers** a, b such that $a \leq S_0 \leq b$ (S_0 is also an integer)
 - a and b form the **barriers** of the RW
- The **hitting time** T of the RW is the first time it touches a or b
 - $T = \min\{n \geq 0 : S_n = a \text{ or } S_n = b\}$





Recap

- **Example 3:**
 - You play a **gambling game**, and your initial position is $0 \leq S_0 \leq 100$
 - You **stop** when $S_n = 0$ (you **bankrupt**) or $S_n = 100$ (you reach your **goal**)
 - ✓ • **Question 1:** will the game end (hit the barrier), or go on for ever?
 - ✓ • **Question 2:** if it ends, what is the probability you will reach your goal?
 - **Question 3:** on average, how long will it take for the game to end?

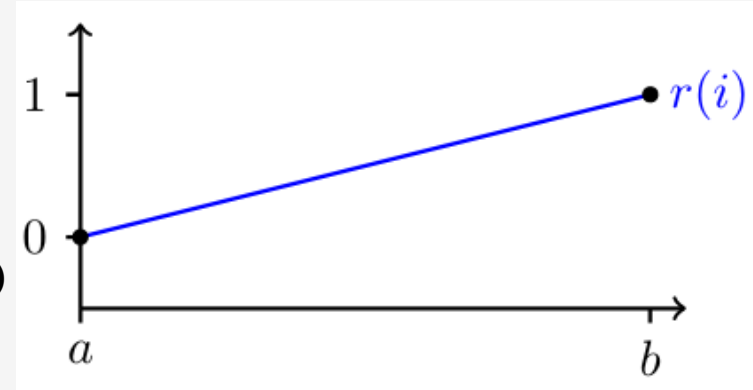
Will be answered in this lecture...

Recap

- Will we ever hit the boundary (e.g., boundary b)?

Question 2:

- $r(i) = P\{S_T = b | S_0 = i\} = \frac{i-a}{b-a}$, $a \leq i \leq b$
(we have two boundaries)



- **Depends on where you start**

- Richer people have more chances to increase their wealth, whereas poorer people have more chances to bankrupt (if $a = 0$) → Philosophical question: **is life fair?**

- The expected \$\$\$ we will have when the game reaches any of the boundaries (if $a = 0, b = 100$)

- $E[S_T | S_0 = i] = 0 \times P\{S_T = 0 | S_0 = i\} + 100P\{S_T = 100 | S_0 = i\} = 100 \frac{i}{100} = i \rightarrow$ **looks fair!**
(mathematically)

Q3: On average, how long will it take for the game to end?

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We are looking for an expectation. But how this expectation looks like ??

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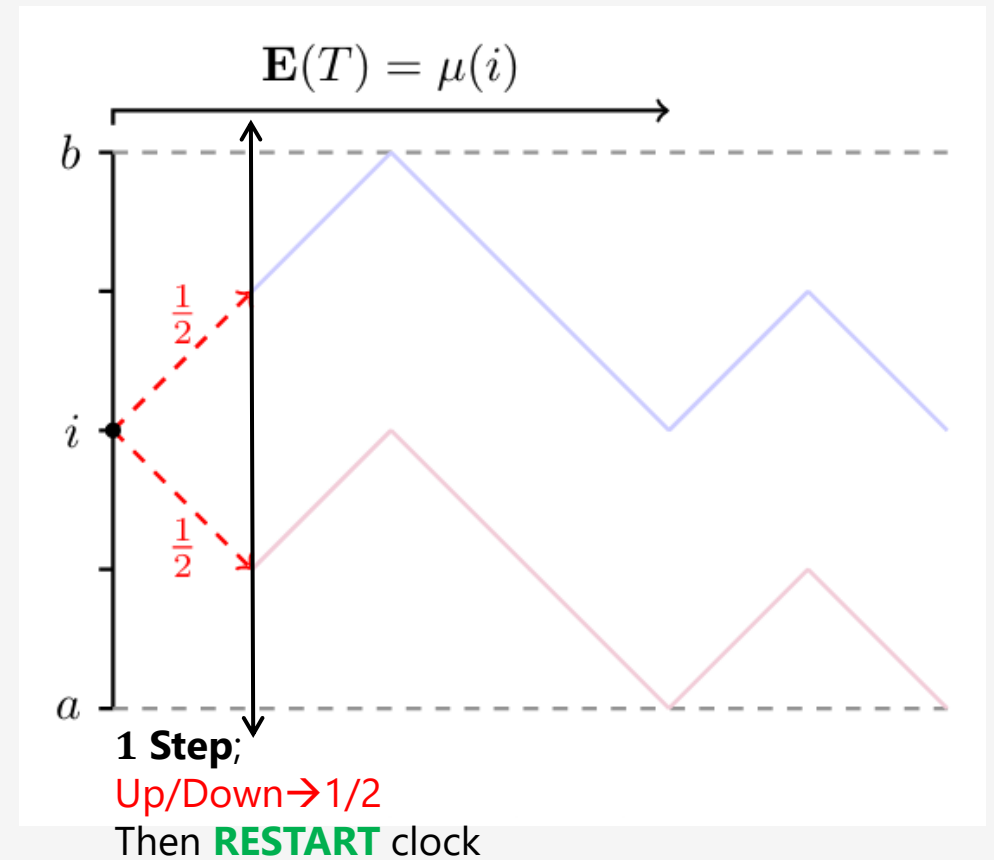
$$\mu(i) = E[T|S_0 = i], \quad a \leq i \leq b$$

- We will also use **First Step Analysis** to answer this question
- **Case #1:** If $i = a$ or $i = b$ then we are already on the boundary, so $T = 0$

$$\mu(a) = E[T|S_0 = a] = 0 \quad \text{and} \quad \mu(b) = E[T|S_0 = b] = 0$$

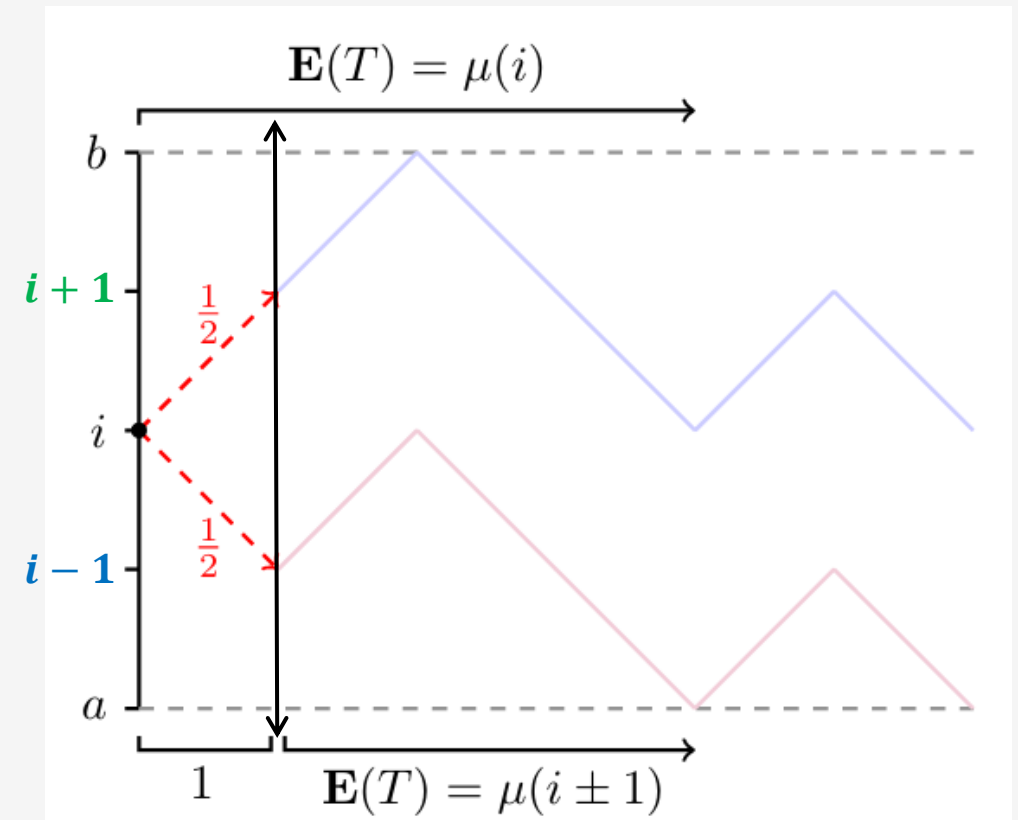
- **Case #2:** If $a < i < b$ by First Step Analysis, we have:

$$\mu(i) = E[T | S_0 = i] =$$



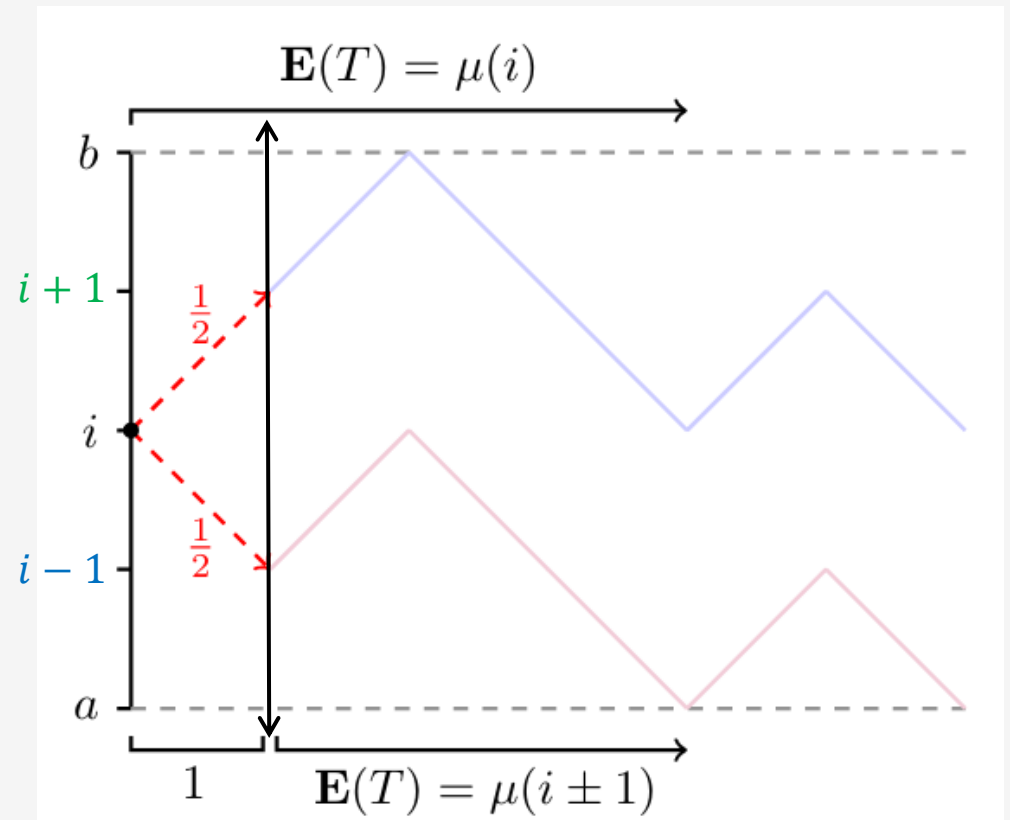
- **Case #2:** If $a < i < b$ by First Step Analysis, we have:

$$\begin{aligned}\mu(i) &= E[T|S_0 = i] = \\ &= (1 + E[T|S_1 = i + 1])P\{S_1 = i + 1|S_0 = i\} + \\ &+ (1 + E[T|S_1 = i - 1])P\{S_1 = i - 1|S_0 = i\} \Rightarrow\end{aligned}$$



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 &+ (1 + E[T|S_1 = i - 1])P\{S_1 = i - 1|S_0 = i\} \Rightarrow \\
 &\Rightarrow \mu(i) = (1 + \mu(i + 1))\frac{1}{2} + (1 + \mu(i - 1))\frac{1}{2} \Rightarrow \\
 &\Rightarrow \mu(i) = 1 + \frac{1}{2}\mu(i + 1) + \frac{1}{2}\mu(i - 1)
 \end{aligned}$$



*** We need to express it in terms of what we know: a, b, i

-
- Hence, we have the system of linear equations:

$$\begin{cases} \mu(a) = 0, & \mu(b) = 0 \end{cases} \quad (1)$$

$$\begin{cases} \mu(i) = \mathbf{1} + \frac{1}{2}\mu(i+1) + \frac{1}{2}\mu(i-1), & a < i < b \end{cases} \quad (2)$$

- The **1** in equation (2) makes the solution **more complicated**

$$(2) \Rightarrow \mu(i+1) - \mu(i) = \mu(i) - \mu(i-1) - \mathbf{2}, \quad a < i < b \quad (2')$$

- We will solve the recursive equations in a **smart way**...

$$(\mu(i + 1) - \mu(i) = \mu(i) - \mu(i - 1) - 2)$$

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... ..

$$i = a + \mathbf{i} - \mathbf{1} \Rightarrow \mu(a + i) - \mu(a + i - 1) = \mu(a + i - 1) - \mu(a + i - 2) - 2 = \dots = \mathbf{u} - (\mathbf{i} - \mathbf{1}) \times 2$$

-
- Summing up all the previous *i* equations (including $\mu(a + 1) - \mu(a) = u$), yields:

$$\mu(a + i) = iu - \sum_{s=1}^i 2(s - 1) = iu - 2 \frac{1}{2} i(i - 1) = iu - i^2 + i = i(u - i + 1) \quad (3)$$

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$$\dots \Rightarrow \mathbf{u} - b + a + 1 = 0 \Rightarrow \mathbf{u} = b - a - 1 \quad (4)$$

-
- Substituting (4) to (3) yields:

$$\mu(a + i) = i(b - a - 1 - i + 1) \Rightarrow \boxed{\mu(a + i) = i(b - a - i)} \quad (3')$$

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$$\begin{aligned} \mu(i) &= \mu(i + a - a) = \mu(a + i - a) = \mu(a + j) \stackrel{\text{use (3')}}{=} j(b - a - j) \Rightarrow \\ \Rightarrow \mu(i) &= (i - a)(b - a - i + a) \Rightarrow \\ &\quad \text{Substitute } j = i - a \end{aligned}$$

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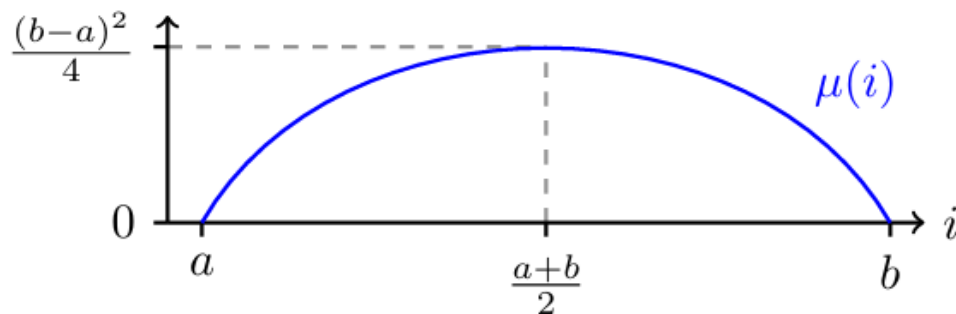
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$\mu(i)$ is a **concave function** with
maximum at $i = \frac{a+b}{2}$ and
maximum value equal to $\frac{(b-a)^2}{4}$

If you start at the middle of the boundary, you have more chances to last longer.

Gambler's ruin



"A Friend in Need (1903)" , by Cassius Marcellus Coolidge
(image credit: Wikipedia)

Gambler's ruin

- Let's now consider a **variation** of our **gambling game**
 - We do **not** have an upper boundary ($b = \infty$) and the lower bound is 0 ($a = 0$)
 - This means that we start with some initial capital of $\$S_0$
 - At every step we **win or lose** \$1
 - If we **bankrupt** ($S_n = \$0$) we **exit** the game
 - **Question:** what is the probability that we will eventually bankrupt?
 - **NOTE, another reasonable question:** is it possible to *keep increasing our wealth indefinitely*?

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 - To use the results we have proved before, we will **convert** our problem to a **two-boundary hitting problem**
 - We define: $T = \min\{n: S_n = 0 \text{ or } S_n = b\}$ for some $b < \infty$
 - But we have proved already that: $T < \infty$ → **Recall that:** $f(i) = P\{T = \infty | S_0 = i\} = 0, \quad a \leq i \leq b$

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 - But we have proved already that: $T < \infty$ → **Recall that:** $f(i) = P\{T = \infty | S_0 = i\} = 0, \quad a \leq i \leq b$
 - Here is a **trick** (to make our life easier):
 - Assume that you **bankrupt in T steps** $\Rightarrow S_T = 0 \xRightarrow{\text{implies}} T_0 = T < \infty$
 - Then $\{S_T = 0\} \xRightarrow{\text{implies}} \{T_0 < \infty\}$

-
- Taking probabilities of these events: $P\{S_T = 0\} \leq P\{T_0 < \infty\} \Rightarrow$

implies

conditioning

$$\Rightarrow P\{T_0 < \infty | S_0 = i\} \geq P\{S_T = 0 | S_0 = i\} =$$

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- Taking $b \rightarrow \infty$ on both sides, we can conclude:
(unlimited funds)

$$\lim_{b \rightarrow \infty} P\{T_0 < \infty | S_0 = i\} \geq \lim_{b \rightarrow \infty} \left(1 - \frac{i}{b}\right) = 1 \quad \Rightarrow \quad P\{T_0 < \infty | S_0 = i\} \geq 1$$

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- But we always have: $0 \leq P\{T_0 < \infty | S_0 = i\} \leq 1$
- Hence: $P\{T_0 < \infty | S_0 = i\} = 1 \rightarrow$ the addicted gambler will eventually lose everything

The gambler will be ruined!

A very important question

- **Question:** On average, how long will it take for the gambler to bankrupt?
- We need to compute: $E[T_0 | S_0 = i] = ??$
- First note that: $T = \min\{n \geq 0 : S_n = 0 \text{ or } S_n = b\} = T_0 \wedge T_b = \min\{T_0, T_b\}$

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 - We start with $\$S_0$
 - We can earn $\$1$ at every step

BUT, what is the minimum number of trials until the gambler reaches b if he started at $S_0 = i$

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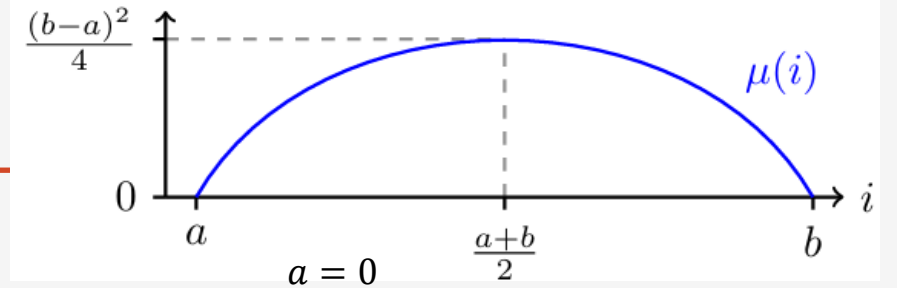
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- But as $b \rightarrow \infty$ we also have $b - S_0 \rightarrow \infty$, and hence $\lim_{b \rightarrow \infty} T_b = \infty$

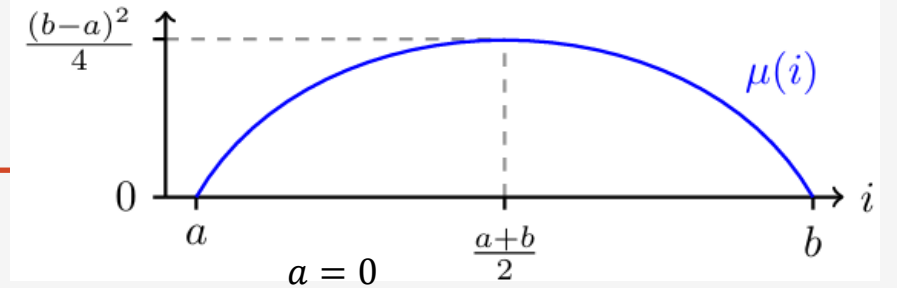
$$\begin{aligned} \Rightarrow \lim_{b \rightarrow \infty} T &= \lim_{b \rightarrow \infty} T_0 \wedge T_b = \\ &= T_0 \wedge \lim_{b \rightarrow \infty} T_b = \\ &= \min\{T_0, \infty\} = T_0 \end{aligned}$$



- Let's now compute the **expected time till we hit bankruptcy**

$$E[T_0 | S_0 = i] = \lim_{b \rightarrow \infty} E[T_0 \wedge T_b | S_0 = i] = \lim_{b \rightarrow \infty} \underbrace{E[T | S_0 = i]}_{\mu(i)} \quad \text{(Slide 27, with } a = 0\text{)} = \lim_{b \rightarrow \infty} i(b - i)$$

- There are **two cases**:

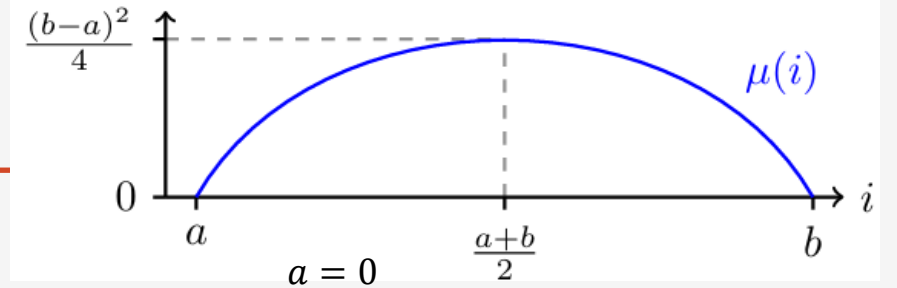


- Let's now compute the **expected time till we hit bankruptcy**

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1. If $i = 0$ then $E[T_0 | S_0 = i] = 0$ (we are already bankrupt!)

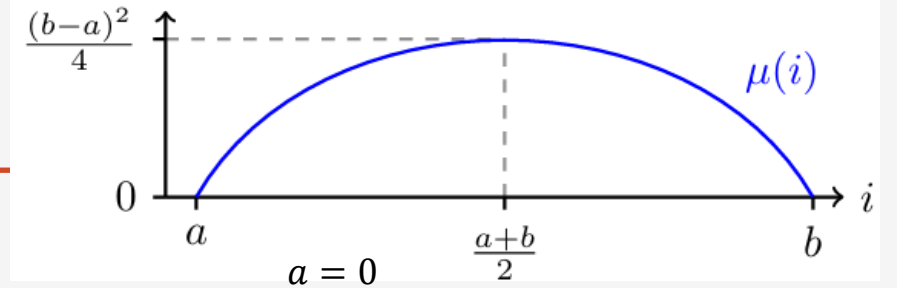


- Let's now compute the **expected time till we hit bankruptcy**

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- There are **two cases**:

- If $i = 0$ then $E[T_0 | S_0 = i] = 0$ (we are already bankrupt!)
- If $i > 0$ then $E[T_0 | S_0 = i] = \lim_{b \rightarrow \infty} i(b - i) = \infty$



- Let's now compute the **expected time till we hit bankruptcy**

$$E[T_0 | S_0 = i] = \lim_{b \rightarrow \infty} E[T_0 \wedge T_b | S_0 = i] = \lim_{b \rightarrow \infty} \underbrace{E[T | S_0 = i]}_{\mu(i)} \quad \text{(Slide 26, with } a = 0\text{)} = \lim_{b \rightarrow \infty} i(b - i)$$

- There are **two cases**:

1. If $i = 0$ then $E[T_0 | S_0 = i] = 0$ (we are already bankrupt!)

2. If $i > 0$ then $E[T_0 | S_0 = i] = \lim_{b \rightarrow \infty} i(b - i) = \infty$

- So, even though **we will bankrupt after finite time**, **the expected time at which we go bankrupt is ∞**

-
- **Some more thinking on this!**
 - $E[T_0|S_0 = i] = \infty$ means that: if we go to **Vegas very many times** (repetition of the experiment) and record the “**number of plays**” we need until we go bankrupt, we will observe that those “**numbers of plays**” are **often very large number**.
 - Hence the average of those “**number of plays**” **goes to** ∞ as we keep going to **Vegas** (and losing all our money).
 - Even though the “**number of plays we need to go bankrupt**” is **finite** (all the time), we **cannot predict the magnitude of that number, before we play**. That’s why its expectation is ∞ .
 - **The catch 22 is:** even though the gambler **will lose in the long run** (if he plays ∞ times), that “long run” is **hard to estimate** and **potentially very long**, giving the false hope that it may be after the gambler’s lifetime.

Doubling strategies

- **Example:**
- Let's go back to **19th century** and play a very risky game (**not advisable!**)
- **You keep playing until you double your initial capital.** If you bankrupt, you borrow money and keep playing. **Note:** $(a, b) = (-\infty, 2i)$
 - **In Math:** you start with $S_0 = \$i$ and you stop gambling only when reach time T_{2i}
- **Question: What is the probability that you will achieve your goal?**
 - We are looking for the probability: $P\{T_{2i} < \infty | S_0 = i\}$
 - Working similarly as before we can compute: $P\{T_{2i} < \infty | S_0 = i\} = 1$,
(the lower boundary now is $-\infty$)

-
- **Working similarly as before ...** we can compute this probability:

$$P\{\mathbf{T}_{2i} < \infty | S_0 = i\} = 1$$

- **But** the expected number of rounds the gambler needs to play before he achieves his “doubling” goal is

$$E[\mathbf{T}_{2i} | S_0 = i] = \infty$$

- So, although the gambler can achieve his “doubling” goal, he needs to play so many rounds, he may not be alive to use the proceeds
 - During the processes he, most likely, will collect a very large debt he may not be able to pay back
